



Master Presenter's Manual

for

Cybersecurity Operations Fundamentals

Curriculum Code: OV-1524

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1. Introduction

This Presenter's Manual is to be used for the **Cybersecurity Operations Fundamentals** course.

After the completion of this course, students will be able to:

- Explain fundamentals of Cybersecurity.
- Illustrate in detail firewall and its features.
- Summarize service models and deployment models of cloud computing.
- Explain Vulnerabilities in Network Infrastructure.
- Explain AAA security model.
- Describe concepts of Cryptography and its features.
- Explain fundamentals of Identity and Access Management (IAM)
- Illustrate protocol analysis and intrusion analysis.
- Define and describe fundamentals of Digital Forensics.
- Illustrate log analysis with NetFlow.
- Explain importance of workflow management along with its components

2. General Information on Session Schedule

The table below shows the details of the modules, the number of hours being spent on the module along with a break-up of the hours being spent on the module.

This course is based on theory and practice lab session. A session is of two hours duration.

Module Wise Session Allocation

Module	Online Hours	Skill Code
Cyber Security Operations Fundamentals	40	1331

Module Deliverables available on OnlineVarsity

To aid the learning process, following are the deliverables

Student Deliverables:

1. Learner's Guide (eBook)

Resources available on OnlineVarsity for Students:

Icons	Feature - Description/Functionality
	Download Book - Student has the option to download the subject related e-book and read offline.
	Glossary - Student can access a list of subject related specialized words with their definitions.
	FAQ - Student can access frequently asked questions and their answers.
	Practice 4 Me - Student can test and evaluate their understanding of module related topics.
	Work Assignments - Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
	References - Student can access additional subject related material for reading.
	Feedback - Student can provide feedback on the course material.
	Ask to Learn - Student can submit subject related technical queries. Queries submitted will be directed to the particular course coordinator/head.

The table below shows the details of the module, the number of hours being spent on the module along with a break-up of the hours being spent on each session.

3. Week Wise Schedule

Topic \ Week		1	2	3	4	5
Cyber Security Operations Fundamentals	D1	TL1	TL5	TL9	TL13	TL17
	D2	TL2	TL6	TL10	TL14	TL18
	D3	TL3	TL7	TL11	TL15	TL19
	D4	TL4	TL8	TL12	TL16	TL20

Where,

D = Day, TL= Online Session

4. Session Coverage

Session No.	Session Title	Session Details	Deliverables' Mapping
1	CBOPS– TL1	All the topics as listed below from Session 1 of <i>Cyber Security Operations Fundamentals</i> book should be covered in this Session. <u>Session 1: Introduction to Cybersecurity</u> ➤ Explain fundamentals of Cybersecurity ➤ Explain Cybersecurity Frameworks ➤ Define common terminologies in Cybersecurity ➤ Illustrate Top 10 OWASP Vulnerabilities	<u>Cyber Security Operations Fundamentals</u> SG - Session 1 XP-Session 1 TG-Session 1
2	CBOPS– TL2	All the topics as listed below from Session 2 of <i>Cyber Security Operations Fundamentals</i> book should be covered in this session. <u>Session 2: Introduction to Cybersecurity Systems</u> ➤ Explain importance of Cybersecurity ➤ Identify types and benefits of Cybersecurity ➤ Illustrate in detail firewall and its features ➤ Summarise Enterprise security architecture	<u>Cyber Security Operations Fundamentals</u> SG - Session 2 XP-Session 2 TG-Session 2
3	CBOPS– TL3	All the topics as listed below from Session 3 of <i>Cyber Security Operations Fundamentals</i> book should be covered in this session. <u>Session 3: Cloud Computing and Cloud Security</u> ➤ Explain cloud computing ➤ Summarise service models and deployment models of cloud computing ➤ Illustrate importance of security in DevOps ➤ Explain difference between Agile and CI/CD	<u>Cyber Security Operations Fundamentals</u> SG - Session 3 XP-Session 3 TG-Session 3
4	CBOPS– TL4	All the topics as listed below from Session 4 of <i>Cyber Security Operations Fundamentals</i> book should be covered in this session. <u>Session 4: Attacks and Vulnerabilities</u> ➤ Explain vulnerabilities in Network Infrastructure ➤ Explain types of cyberattacks ➤ Define footprinting and Reconnaissance	<u>Cyber Security Operations Fundamentals</u> SG - Session 3 XP-Session 3 TG-Session 3

Session No.	Session Title	Session Details	Deliverables' Mapping
5	CBOPS– TL5	All the topics as listed below from Session 5 of Cyber Security Operations Fundamentals book should be covered in this session. <u>Session 5: Access Control Models and Methods</u> ➤ Explain fundamentals of access control ➤ Explain AAA security model ➤ Illustrate various types of access control ➤ Summarise antivirus and anti-malware solutions	<u>Cyber Security Operations Fundamentals</u> SG - Session 5 XP-Session 5 TG-Session 5
6	CBOPS– TL6	All the topics as listed below from Session 6 of Cyber Security Operations Fundamentals book should be covered in this session. <u>Session 6: Overview of Cryptography and PKI</u> ➤ Explain concepts of cryptography and its features ➤ Distinguish between encryption and decryption ➤ Explain algorithms in cryptography ➤ Illustrate the process of hashing and its features	<u>Cyber Security Operations Fundamentals</u> SG - Session 6 XP-Session 6 TG-Session 6
7	CBOPS– TL7	All the topics as listed below from Session 7 of Cyber Security Operations Fundamentals book should be covered in this session. <u>Session 7: Introduction to VPN</u> ➤ Explain VPN and its implementations ➤ Outline security features of VPN ➤ Illustrate protocols for configuring VPN	<u>Cyber Security Operations Fundamentals</u> SG - Session 7 XP-Session 7 TG-Session 7
8	CBOPS– TL8	All the topics as listed below from Session 8 of Cyber Security Operations Fundamentals book should be covered in this session. <u>Session 8: Security Operations and Management</u> ➤ Explain fundamentals of Identity and Access Management (IAM) ➤ Identify Risks and challenges in IAM ➤ Define IAM Life Cycle ➤ Define and describe asset management and configuration management	<u>Cyber Security Operations Fundamentals</u> SG - Session 8 XP-Session 8 TG-Session 8

Session No.	Session Title	Session Details	Deliverables' Mapping
9	CBOPS– TL9	All the topics as listed below from Session 9 of Cyber Security Operations Fundamentals book should be covered in this session. <u>Session 9: Overview of Intrusion Analysis</u> ➤ Explain fundamentals of incident response ➤ Explain incident response team structure ➤ Define Managed Security Services Providers ➤ Illustrate protocol analysis and intrusion analysis	<u>Cyber Security Operations Fundamentals</u> SG - Session 9 XP-Session 9 TG-Session 9
10	CBOPS– TL10	The following topics as listed below from Session 10 of Cyber Security Operations Fundamentals book should be covered in this session. <u>Session 10: Digital Forensics</u> ➤ Define and describe fundamentals of Digital Forensics ➤ Explain types of Digital Forensics	<u>Cyber Security Operations Fundamentals</u> SG - Session 10 XP-Session 10 TG-Session 10
11	CBOPS– TL11	The following topics as listed below from Session 10 of Cyber Security Operations Fundamentals book should be covered in this session. <u>Session 10: Digital Forensics</u> ➤ Illustrate performing forensics on Windows ➤ Illustrate performing forensics on Linux Operating System	<u>Cyber Security Operations Fundamentals</u> SG - Session 10 XP-Session 10 TG-Session 10
12	CBOPS– TL12	The following topics as listed below from Session 11 of Cyber Security Operations Fundamentals book should be covered in this session. <u>Session 11: Log Analysis</u> ➤ Explain fundamentals and benefits of logs ➤ Describe packet capturing	<u>Cyber Security Operations Fundamentals</u> SG - Session 11 XP-Session 11 TG-Session 11
13	CBOPS– TL13	The following topics as listed below from Session 11 of Cyber Security Operations Fundamentals book should be covered in this session. <u>Session 11: Log Analysis</u> ➤ Illustrate log analysis with NetFlow ➤ Summarise the 5-Tuple Correlation Mode	<u>Cyber Security Operations Fundamentals</u> SG - Session 11 XP-Session 11 TG-Session 11

Session No.	Session Title	Session Details	Deliverables' Mapping
14	CBOPS– TL14	The following topics as listed below from Session 12 of Cyber Security Operations Fundamentals book should be covered in this session. <u>Session 12: Challenges in Security Operations Center (SOC)</u> ➤ Explain fundamentals of security monitoring ➤ Explain importance of NTP in SOC environment	<u>Cyber Security Operations Fundamentals</u> SG - Session 12 XP-Session 12 TG-Session 12
15	CBOPS– TL5	The following topics as listed below from Session 12 of Cyber Security Operations Fundamentals book should be covered in this session. <u>Session 12: Challenges in Security Operations Center (SOC)</u> ➤ Define DNS tunnelling and peer-to-peer communication ➤ Illustrate various evasion techniques	<u>Cyber Security Operations Fundamentals</u> SG - Session 12 XP-Session 12 TG-Session 12
16	CBOPS– TL6	The following topics as listed below from Session 13 of Cyber Security Operations Fundamentals book should be covered in this session. <u>Session 13: Intrusion Events</u> ➤ Explain fundamentals of Intrusion Events ➤ Explain Diamond Model of Intrusion Event	<u>Cyber Security Operations Fundamentals</u> SG - Session 13 XP-Session 13 TG-Session 13
17	CBOPS– TL17	The following topics as listed below from Session 13 of Cyber Security Operations Fundamentals book should be covered in this session. <u>Session 13: Intrusion Events</u> ➤ Define Cyber Kill Chain Model ➤ Summarise Secure Malware Analytics	<u>Cyber Security Operations Fundamentals</u> SG - Session 13 XP-Session 13 TG-Session 13
18	CBOPS– TL18	All the topics as listed below from Session 14 of Cyber Security Operations Fundamentals book should be covered in this session. <u>Session 14: Threat Hunting</u> ➤ Explain fundamentals of threat hunting ➤ List various tools and models of threat hunting ➤ Define threat intelligence	<u>Cyber Security Operations Fundamentals</u> SG-Session 14 XP-Session 14 TG-Session 14

Session No.	Session Title	Session Details	Deliverables' Mapping
19	CBOPS– TL19	The following topics as listed below from Session 15 of <i>Cyber Security Operations Fundamentals</i> book should be covered in this session. Session 15: Workflow Management in SOC ➤ Explain workflow management ➤ Explain importance of workflow management along with its components	<u>Cyber Security Operations Fundamentals</u> SG-Session 15 XP-Session 15 TG-Session 15
20	CBOPS– TL20	The following topics as listed below from Session 15 of <i>Cyber Security Operations Fundamentals</i> book should be covered in this session. Session 15: Workflow Management in SOC ➤ List key benefits of SOC Automation	<u>Cyber Security Operations Fundamentals</u> SG - Session 15 XP-Session 15 TG-Session 15

5. Deliverables

The deliverables to be provided to the students and the faculty have been tabulated below:

➤ Deliverables for Students:

Learner's Guide	eBook consisting of the knowledge to be imparted (wherever applicable)
Assignments	Case study/scenario based problems to be solved by learners
Article	Additional knowledge in the form of articles that are referred from Web
Practice Questions	Additional practice questions for learners at the end of each session

Module	Book Code	Book Name
KIT	OV-1524-KIT01-01-A	Kit of 1 title for OV-1524 Cybersecurity Operation
Cybersecurity Operations Fundamentals	AINNCYOPS10423E000	Cyber Security Operations Fundamentals

➤ **Deliverables for Faculty**

Presenter's Manual

A training delivery plan for the trainers

Trainer's Guide

A session wise guide for the trainers

Classroom
Presentations

Session wise PowerPoint presentations as
teaching aid, wherever applicable

Module Name	Book Code	Book Name
KIT	OV-1524-KIT01-01-A	Kit of 1 title for OV-1524 Cybersecurity Operation
Cybersecurity Operations Fundamentals	AINNCYOPS10423E000	Cyber Security Operations Fundamentals

6. Library References

- **Understanding Cisco Cybersecurity Operations Fundamentals (CBROPS 200-201) Exam Practice Tests & Dumps** by Books Fortune
- **Introduction to Cyber Security: Guide to the World of Cyber Security** by Anand Shinde

7. Evaluation Strategy

The Evaluation Strategy for this course is as follows:

- Final Examination (Objective)
- Practical Examination

7.1 Award of Grades

Overall Weighted Marks > = 40% but < 60% qualifies for **PASS**

Overall Weighted Marks > = 60% but < 75% qualifies for **CREDIT** and

Overall Weighted Marks > = 75% qualifies for **DISTINCTION**

Note: To attain a PASS/CREDIT/DISTINCTION grade, a student should achieve at least 40% in each of the elements of the Examination; otherwise he/she will be declared as 'Referred'.

7.2 Final Examination 1 (E1)

Below is the list of modules to be covered in Final Examination along with their weightages:

Final Examination 1 (E1)

Marks: 50

Duration: 1 hour

Type: Objective

S#	Name of the Subject	Marks
1.	Cybersecurity Operations Fundamentals	50

Final Examination 2 (R1)

Marks: 50

Duration: 2 hours

Type: Practical

S#	Name of the Subject	Marks
1.	Cybersecurity Operations Fundamentals	50

Evaluation Details

OES-50%

Practical-50%

8. Classroom and Lab Pre-requisites

The requirements for conducting the sessions have been described as follows:

8.1 Software Requirements

Microsoft Windows 11

Graphical Network Simulator 3(GNS3) or any other Router Simulation Software

8.2 Recommended Hardware

Intel Core i5/i7 Processor or higher

8 GB or 16 GB RAM

1 TB Hard Disk space

Mouse

Keyboard

8.3 Other Requirements (For Classroom)

Computer for giving demonstrations wherever applicable

LCD projector

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